Project Report: A Unified Approach to Interpreting Model Prediction

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1. Project Overview

The objective of this project is to implement and validate the framework proposed in the research paper **"A Unified Approach to Interpreting Model Predictions"**. This work introduces **SHAP (SHapley Additive exPlanations)**, a unified framework for interpreting model predictions that assigns each feature an importance value for a specific prediction.

The implementation focuses on reproducing the paper's approach to explaining complex model outputs—specifically a **Convolutional Neural Network (CNN)** trained on the **MNIST digit dataset**—and comparing SHAP with another popular interpretability method, **LIME**.

2. Theoretical Background: The SHAP Framework

SHAP is built upon the identification of a new class of **additive feature importance measures**. The paper demonstrates that several existing methods, including **LIME**, **DeepLIFT**, and **Layer-Wise Relevance Propagation**, all fall into this class.

Key Properties of SHAP

The framework is mathematically grounded in cooperative game theory, proving that there is a unique solution in the class of additive feature attribution methods that satisfies three desirable properties:

- **Local Accuracy:** The sum of the feature attributions matches the original model output for a specific input.
- **Missingness:** Features that are missing in the original input are constrained to have no attributed impact.
- **Consistency:** If a model changes so that a feature's contribution increases, its attribution should not decrease.

3. Implementation Details

The project was implemented in Python using **TensorFlow/Keras** for model development and the `shap` and `lime` libraries for interpretability.

Model Architecture

A CNN was constructed with the following layers to classify MNIST digits:

- **Input Layer:** $28 \times 28 \times 1$ grayscale images.
- **Convolutional Layers:** Two layers with 3×3 kernels and ReLU activation.
- **Pooling Layers:** Max pooling (2×2) for spatial reduction.
- **Dense Layers:** A fully connected layer with 128 units and a final 10-way **Softmax** output layer.

Interpretability Implementation

- **SHAP:** Utilized the `GradientExplainer` to calculate SHAP values, establishing a baseline using 100 random background images from the training set.
- **LIME:** Used the `LimeImageExplainer` with 1,000 perturbations to identify the top 5 features (super-pixels) contributing to the prediction.

4. Results and Analysis

The model achieved high performance quickly, reaching a **validation accuracy of 98.62%** after just two epochs.

Visual Comparison

As seen in the generated output for the digit '7', both methods were evaluated based on their ability to highlight relevant features:

- **SHAP Explanation:** Provides a granular, pixel-level heat map showing both positive (red) and negative (blue) contributions to the class probability.
- **LIME Explanation:** Highlights broader "super-pixel" regions that influenced the model's decision.

Performance Insights

The implementation confirms the paper's findings that SHAP offers a more consistent and theoretically grounded explanation compared to LIME. While LIME is model-agnostic and flexible, it relies on heuristic choices for its local kernel, which can lead to violations of local accuracy or consistency. SHAP, conversely, provides a unique solution that better aligns with human intuition.

5. Conclusion

This project successfully reproduced the core methodology of the SHAP framework as applied to deep learning. By unifying disparate interpretation methods under the umbrella of additive feature attribution, SHAP provides a rigorous standard for Explainable AI (XAI). Future work could involve applying **Deep SHAP** to leverage the compositional nature of deeper networks for even faster computation.